Andrey Mazépas

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Summary

Software Engineer with over a decade of experience working on web and native applications, specializing in real-time 3D visualization, spatial data and performance-critical product interfaces.

Experienced in building and owning complex frontend architectures across robotics, construction mapping, financial services and SaaS industries.

Skills

React, React Native, Three.js, React Three Fiber, TypeScript, JavaScript, Next.js, C++, OpenGL, HTML, CSS, Python, Material UI, Vite

Experience

Red Dot Drone Japan, Inc. | Osaka, Japan
Native Application Engineer | 07/2025 - Present

- Lead development of real-time 3D visualization and measurement systems for drone-acquired point-cloud data used in civil construction and mapping, using React, Three.js (React Three Fiber), C++ and OpenGL.
- Develop and evaluate scalable rendering approaches, including COPC-based web visualization, with a target of up to 50 million points on lightweight 8 GB machines.
- Design rendering pipelines for memory efficiency and responsive interaction; contribute to drone-system simulation and hardware integration workflows. Supported live remote-drone demonstrations at Expo 2025 Osaka.

Formant, Inc | Mill Valley, US (Remote)
Software Engineer (Full-time Independent Contractor) | 10/2022 - 04/2025
Cloud-based robotics fleet telemetry platform

- Maintained and extended a modular React and TypeScript frontend for real-time robot telemetry, contributing to multi-robot “Coherence views” with shared timeline synchronization and user customization.
- Owned the React Three Fiber / Three.js 3D Scene Module: transform-synchronized overlays, dynamic maps, live URDF and joint-state rendering, mission-planning layers, and context-aware camera behavior.
- Built a custom ROS-marker rendering system using Three.js instancing, supporting updates to lists of 100,000 markers while rendering at 60 fps on a standard MacBook Air.
- Engineered image/video modules for browsing 100+ feeds in a virtualized, quickly scrollable view, using smart prefetching while supporting frame-accurate playback and fast seeking.

Divinia | San Francisco, US (Remote)
Software Engineer (Full-time Independent Contractor) | 11/2021 - 10/2022
AI-based product categorization and search

- Led frontend development for 3 SEO-optimized product search platforms using React and Next.js. Partnered across product work and Kotlin backend services to deliver end-to-end features.

RunaHR | Mexico City, MX (Remote)
Software Engineer (Dispatched via Turing) | 08/2021 - 11/2021
Payroll management SaaS

- Delivered payroll automation interfaces with React and Redux while collaborating in Agile sprints across distributed engineering teams.

Zenvia | São Paulo, BR (Remote)
Frontend Developer (Full-time Employee) | 01/2021 - 08/2021
Enterprise messaging and communication solutions

- Built and maintained React/Redux chat, administration and dashboard applications; developed a performance-critical embeddable JavaScript chat widget and custom voice-recorder functionality.

Banco do Brasil | Brasília, BR (On-site)
Frontend Developer (Dispatched via EWave) | 10/2019 - 01/2021
Internal systems for Brazil's largest bank

- Built internal dashboards and charting tools using AngularJS integrated with Java APIs.
- Maintained Java endpoints for frontend requirements and delivered pension features for the official banking app with React Native within a security and compliance-focused environment.

Output Sistemas | Brasília, BR (On-site)
Fullstack Developer (Full-time Independent Contractor) | 05/2019 - 10/2019
Fintech startup

- Created a business asset management desktop app using React and Electron, owning frontend and backend logic.
- Designed and implemented a corporate landing page with React.

Various | Brasília, BR (On-site)
Software Developer (Consolidated Early Career Roles) | 06/2015 - 05/2018

- Built and maintained web platforms for public institutions and research groups using Angular and React.
- Delivered a live data broadcasting web system for the Brazilian Army, deployed to military bases nationwide.
- Created React interfaces for federal talent-management systems with Material UI and Ant Design, with Node.js API integration; also supported departmental IT and server administration.

Languages

English, Japanese (JLPT N2 Certification), Portuguese, Spanish

Education

University of Brasilia | Brasília
Computer Science | 01/2021
Completed 3 years of the bachelors program.